Programming and production

The labs and practice sets of Day 5. You type your work in the do-file named on each page; this booklet is the printed copy to read, plan and write notes on. Core first, then Extension, then Challenge.

How the checks work. Every task in the do-file ends with a CHECK line. Run the file with Ctrl + D: a silent CHECK means your answer is correct; red text under a CHECK means look at that task again. Never edit a CHECK line. The solutions booklet explains the reasoning behind each answer; read it after you have tried.

Lab 1: Assemble the pipeline

File `C:/NBS_Stata/day5/labs/lab_D5_1.do`
inputs data/lfs_raw.dta, data/cpi_divisions_long.dta,
outputs logs/lab_D5_1.log and everything the modules write

The situation
Next quarter a colleague must produce the LFS mini-release while you are on leave. She should change one block of settings, press Ctrl + D, and get the same files you get today. This lab file is that master do-file.

Task 1 The settings block — Core

version 17, clear all, and three globals: root (C:/NBS_Stata), quarter (2026Q1) and youth_max (24).

Task 2 Check the inputs before anything runs — Core

A foreach loop over the two input files (data/lfs_raw.dta and data/cpi_divisions_long.dta). For each: capture confirm file; if it fails, display a red message naming the file and exit 601.

Task 3 Folders and one log — Core

Create the folders output and logs if they are missing (capture mkdir). Close any open log, then open logs/lab_D5_1.log (replace, text).

Task 4 The five modules, timed — Core

timer clear and timer on 1; run the five modules in order with do; timer off 1 and timer list 1.

Task 5 Read the result back — Core

Import the Regions sheet of output/estimates_2026Q1.xlsx (firstrow, clear) and list it. Which region has the highest rate?

Task 6 One person, one row — Extension

(a) Open the raw file and run duplicates report unique_ind: how many surplus rows are there? (b) Open data/lfs_2026Q1_teach.dta. Check it with capture isid unique_ind; if that fails, display a red message and exit 459.

Task 7 Which module is the slowest? — Challenge

Put the five module names in a local. In a foreach loop, give each module its own timer (1 to 5): timer on, quietly do, timer off. Then timer list. Close the log at the end.

A correct lab. the whole file runs with no red text and ends with ”end of do-file”.

Practice sets

File `C:/NBS_Stata/day5/practice/practice_D5.do`. One block per part of the day; each opens its own data, so the parts can be done in any order. Type your number after `nbscheck` where the file asks for it.

Part 1: Macros (lfs_teach.dta)

P1.1 Core Store 15 in a local lo and 24 in a local hi. Count the persons aged lo to hi with inrange(). Type the number.

P1.2 Core A local wk = 48. Count the employed (ilo_lfs19 == 1) whose usual hours are above wk. Type the number.

P1.3 Core How many variables does the file in memory have? Use c(k).

P1.4 Extension Open data/lfs_raw.dta with a path built from the global root. Type its number of observations (c(N)).

P1.5 Challenge local a 2+2 holds text, not a number. What does display `a' * 3 show? Work it out first, then run it.

Part 2: Loops (lfs_svy.dta, four flags)

P2.1 Core A forvalues loop over the 7 regions that counts the labour force in each. Type the count for EastSouth (4).

P2.2 Core Design-based participation rate (lf, subpop wap and the region) in a loop. Type Praslin & La Digue (6), percent, one decimal.

P2.3 Core foreach over the numbers 15 25 35 55: count the employed aged that and over. Type the count for 55.

P2.4 Extension foreach over the variables lf unemp emp: the design-based mean of each in the working-age population (subpop wap). Type the share employed, percent, two decimals.

P2.5 Challenge Participation by region and sex, a loop inside a loop. Type the rate for women (2) in West (5), percent, one decimal.

Part 3: Modules (the three settings are set)

P3.1 Core Run module 01 with do. How many persons are in the file it leaves in memory? (c(N))

P3.2 Core Run module 02. How many young people are in the labour force (lf_youth == 1)?

P3.3 Core Without saving anything: in the file module 02 left in memory, count the labour force aged 15 to 19.

P3.4 Extension Run module 03, then import its National sheet. Type the youth unemployment rate, two decimals.

P3.5 Challenge Set the quarter to 2025Q4 and run module 03 under capture. Type the return code. Then set the quarter back to 2026Q1.

Part 4: The master do-file (run master.do or Lab 1 first)

P4.1 Core Import the Regions sheet of output/estimates_2026Q1.xlsx. Type the rate of Central 1, two decimals.

P4.2 Core Confirm that four products of master.do exist: the two Excel files, the margins chart and the release.

P4.3 Core capture mkdir the output folder, which exists. Keep the return code in a local and type it.

P4.4 Extension Import the Margins sheet of output/model_2026Q1.xlsx. Type the predicted rate for 55+, two decimals.

P4.5 Challenge Import the CPI sheet of the same file. Type the inflation figure, two decimals, and assert the month is August 2026.

Part 5: The release (National sheet in memory; close Word)

P5.1 Core Display the participation rate as text with one decimal (the : display function, format %3.1f). Type it.

P5.2 Core A Word file output/p5_d5.docx: a title with $quarter and one sentence giving the national unemployment rate.

P5.3 Core A Word file output/p5_table.docx with a 4 x 2 table: a header row, then one row per indicator, rate with one decimal.

P5.4 Extension A Word file output/p5_chart.docx: the inflation chart (output/infl_2026Q1.png) and one sentence with the month and the rate, read from the CPI sheet.

P5.5 Challenge Without changing any file: the design-based youth unemployment rate for ages 15 to 29 (proportion), percent, two decimals.

Part 6: Refactoring (lfs_raw.dta, duplicates dropped)

P6.1 Core Drop persons under 15 and build lf and un as the legacy file does, one line each. Type the weighted unemployment rate (summarize with aweight = weights_dis), percent, two decimals.

P6.2 Core The same rate for women (dem_sex == 2), two decimals.

P6.3 Core The same rate for Praslin & La Digue (region_q == 6), two decimals.

P6.4 Extension How many persons did drop if ilo_age < 15 remove? Reopen the raw file, drop the duplicates, and count.

P6.5 Challenge The legacy file builds lf in two lines (gen ... if, then replace). Build it that way and in one line (lf2), and assert that the two agree.

Part 7: Discipline (lfs_2026Q1_svy.dta)

P7.1 Core Which version does this file run under? Type c(version).

P7.2 Core datasignature set, reset; then capture datasignature confirm. Keep _rc in a local and type it.

P7.3 Core Change ilo_age in row 1, then capture datasignature confirm again. Type the return code.

P7.4 Extension Open data/lfs_raw.dta and save it without replace, under capture. Type the return code. (Nothing is overwritten.)

P7.5 Challenge How many do-files are in day5/modules? Put the list in a local with the dir function, then count its words.